Prerak Bhandari

Raleigh, NC | +1 (919) 420-4824 | pbhanda4@ncsu.edu | linkedin.com/in/prerak-bhandari | github.com/bhandariprerak | Portfolio

EDUCATION

M.S. Computer Science - North Carolina State University, Raleigh August 2024 - May 2026
Coursework: Design and Analysis of Algorithms, Computer Networks, DevOps, OOD. ***GPA: 4.00/4.00***

B.E. Computer Science - R. N. S. Institute of Technology, India July 2018 - August 2022
Coursework: AI/ML, Operating Systems, Distributed Systems. ***GPA: 9.02/10.00***

TECHNICAL SKILLS

Programming Languages: Go, Python, Java, C, JavaScript, TypeScript, Ruby, SQL, HTML, CSS, PHP.
Frameworks: Spring Boot, React.js, Node.js, Django, Flask, FastAPI, Ruby on Rails
Databases: MongoDB, MySQL, PostgreSQL, Redis, Elasticsearch, DynamoDB.
Tools: AWS (EKS, S3, EC2, Lambda, ELB, SQS), Azure, Docker, Kubernetes, Ansible, Git, RabbitMQ, Jira, Jenkins

WORK EXPERIENCE

Graduate Student Developer - OSS(Expertiza) January 2026 - Present
North Carolina State University *Raleigh, NC*

- Designed and engineered backend features for **Expertiza OSS (Ruby on Rails, TypeScript)** supporting scalable peer review workflows, rubric-based grading, and submission evaluation pipelines.
- Refactored legacy grading models into a reusable hierarchy, reducing code duplication by **30%**, and **enhanced submission response APIs** to improve peer review workflow reliability and backend testability.

Software Engineer Intern May 2025 - August 2025
Tailon Labs *Remote (USA)*

- Developed a **full-stack Real Estate Management** platform using **React.js, Node.js, TypeScript, and PostgreSQL**, enabling realtors to manage property listings at scale, and **improving operational efficiency by 35%**.
- Optimized **RESTful API** architecture with asynchronous I/O, connection pooling, and Redis-based caching, achieved **3 times throughput** and **reduced p95 latency from 280 ms to 85 ms** under load.

Software Engineer April 2022 - July 2024
Kaleyra Inc. *Bangalore, India*

- Owned and scaled **SMS pipelines** to handle **millions of messages**; overhauled **Sender ID, template, and content APIs**, improved message delivery by **20%** and campaign performance.
- Engineered **multi-channel messaging (Failover)** across SMS, WhatsApp, RCS, Voice, Email using AWS SQS based failover routing to achieve **99.9% reliability** and reduced delivery failures by **40%.**
- Built **WhatsApp capabilities** with **Meta Graph API** (Tech Provider, Account Sharing), automated billing and eliminated billing errors.
- Led implementation of custom domain routing and analytics tracking for the URL Shortener platform across a **3 engineer team**, powering enterprise messaging workloads reaching **5M+ end users**.
- Migrated **150+ APIs** from PHP to **Go** using **gRPC**, improved **latency by 97.5%** and scaled **throughput from 4000 to 15,000 requests per second** while moving the platform from monolith to **microservices**.
- Developed real-time campaign tools including an **Excel SMS Add-in** which improved workflow efficiency by **25%**, and contributed to an **18%** revenue increase.

PROJECTS

Sword URL Shortener (swrd.shop) (Frontend Source Code) (Backend Source Code)

- Built a **99.9% uptime** production-grade URL Shortener using **Java (Spring Boot), PostgreSQL, and React.js**; scaled to handle **1000+ RPS** with JWT authentication, real-time analytics, and **RabbitMQ**-based background processing.
- Built an end-to-end **CI/CD pipeline** running automated tests and linting, packaging services into **Docker** images, and deploying to **AWS EC2** via **Ansible** with controlled release flows from develop to main.

Smart Inventory AI Bot (Source Code)

- Developed a cross-platform, **LLM-integrated mobile app (React Native, FastAPI, Google Gemini AI)** that connects to Google Drive CSVs for real-time inventory, customer, and pricing queries.
- Built a structured query analysis and conversational response pipeline, **reducing query-to-answer latency by 40%**.
- Integrated Google Drive API with caching (pandas DataFrames) for **live data insights across 5000+ records**.

PUBLICATION

- **Prerak Bhandari**, Dr. E. F. Gehringer *"LLM Conversation Transcripts in Software Engineering: Redefining Homework in the Age of Generative AI."* 2026 American Society for Engineering Education (ASEE) Annual Conference and Exhibition, Charlotte, NC